

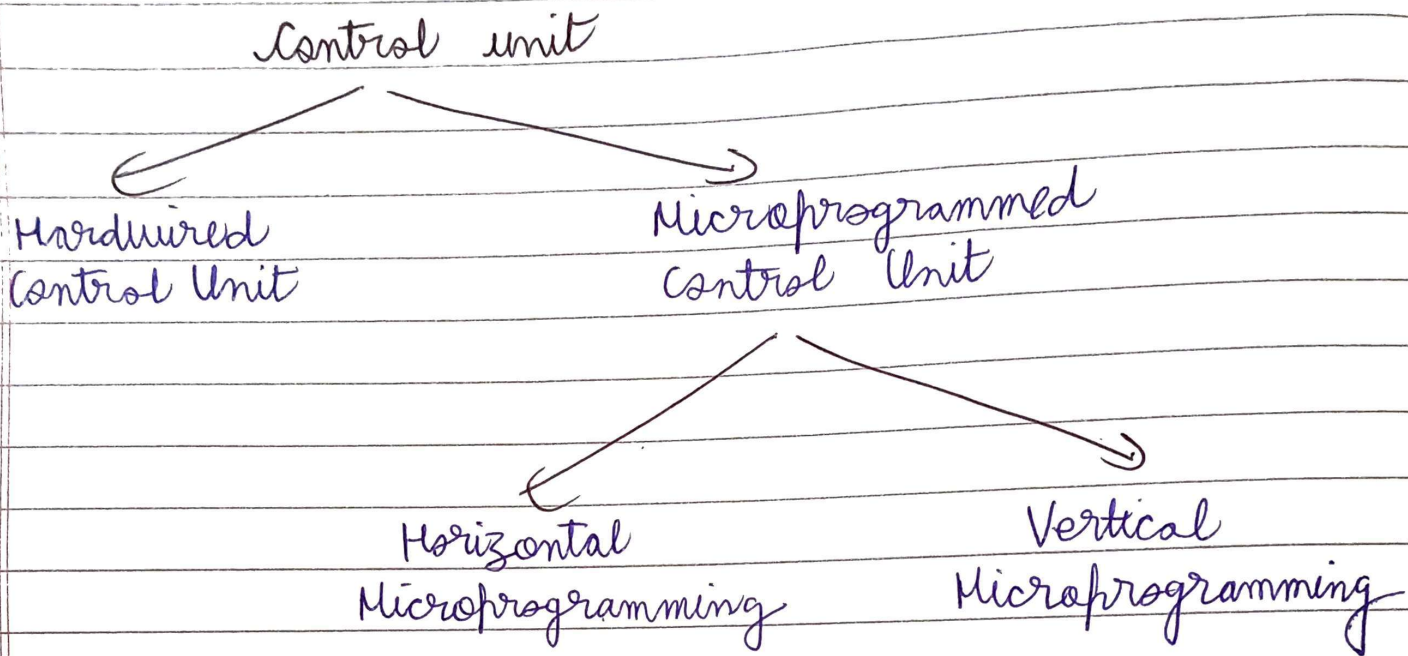
Direct Memory Access (DMA)

- It is designed by Intel to transfer data at the fastest rate.
- It allows the device to transfer data directly to/ from memory without CPU involvement.

How DMA operations are performed?

- ① Device (Peripheral) wants to send data to memory.
Firstly, device has to send DMA request (DRQ) to DMA controller.
- ② DMA controller sends Hold request (HLQ) to CPU. and waits for the CPU to send HLQ acknowledgement (HLDA).
- ③ The CPU leaves the control over bus and acknowledge the HLDA signal to DMA controller.
- ④ Now, CPU is in the HOLD state, DMA controller has to manage operations over buses. w/w CPU, memory and I/O devices.

By Me:



- ① Each word in control memory contains within it a microinstruction. The microinstruction specifies one or more microoperations for the system.
- ② A sequence of Microinstruction is called a Microprogram.

Hardwired Control Unit

Microprogrammed Control Unit

- | | | |
|----|---|--|
| 1. | Speed: Very fast (single-cycle microoperations) | Speed: Slower than hardwired (multiple memory accesses). |
| 2. | Flexibility: Difficult to modify or extend once designed. | Flexibility: Easy to modify control sequences. |
| 3. | Complexity: Logic grows exponentially | Complexity: Suited for complex ISAs CCISC |

4.

with ISA components.

Use cases: Simple RISC
processors

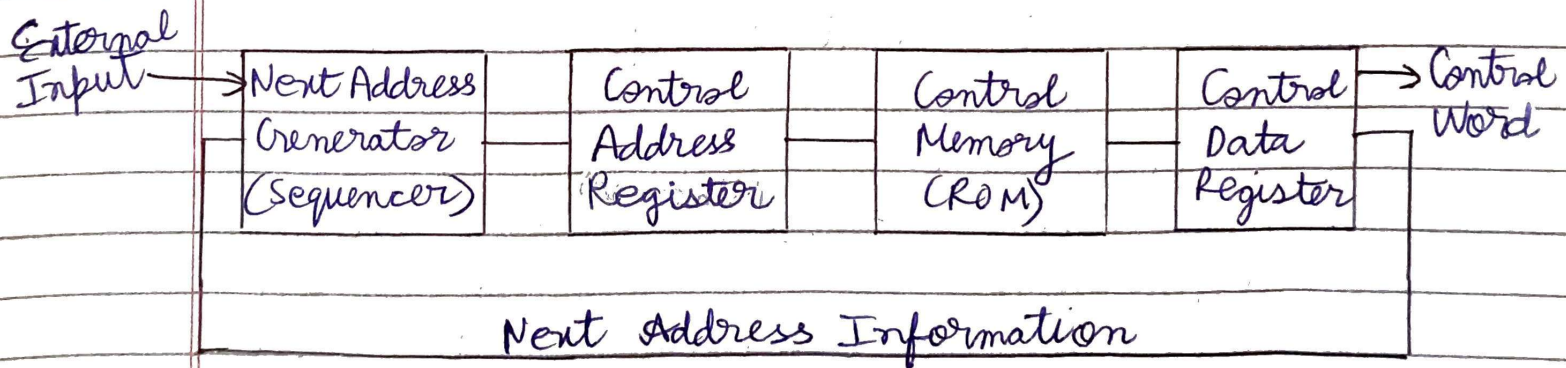
with limited instruction
sets.

with many instructions.

Storage: Requires ROM/RAM
for control memory
(2K - 10K microinstructions.)

MICROPROGRAMMED CONTROL ORGANISATION

- Computer contains three components: CPU, I/O Devices, Memory.
- CPU contains three components: Registers, ALU, Control Unit
- Microprogram is a sequence of microinstructions and a microinstruction contains one or more microoperation.
- Control Address Register holds the address of instruction.
- Control Data Register contains the data of the next instruction.
- Next Address Generator will generate the next instruction address that is to be executed
- Control Data Register is also called as Pipelining register means executing multiple task simultaneously in less amount of time.



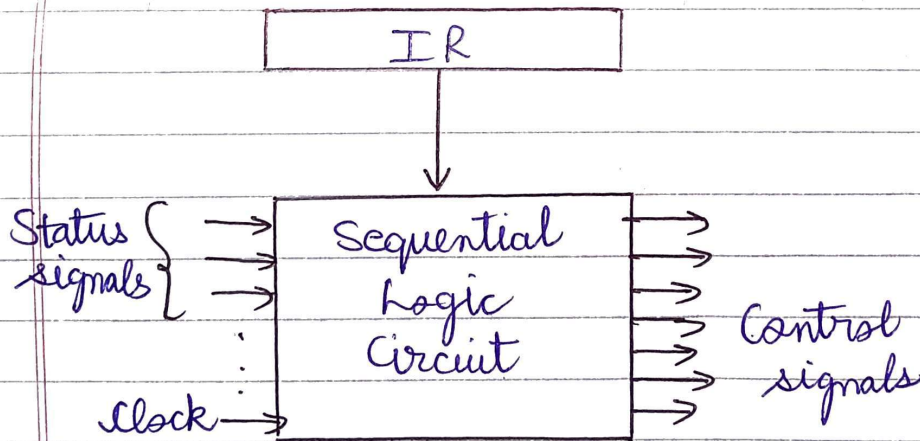
HARDWIRED CONTROL UNIT

A control unit can be implemented basically by two techniques:

1. Hardwired implementation
2. Micro-programmed implementation

Hardwired control unit is implemented as a sequential logic circuit or a finite state machine that generates a specific sequence of control signals.

Structure



- It uses fixed logic.
- Interprets & then generates corresponding control signals.
- Factors for design:
 - Amount of hardware
 - Speed for operation
 - Cost

→ There are four techniques for design of hardwired control unit.

1. State-table method:

- ⊙ Classical method of sequential design
- ⊙ Minimizes hardware
- ⊙ Constructs a state transition table.
- ⊙ Every generation of states has a set of control signals.

2. Delay - element method:

- ⊙ Control signals follow a proper sequence.
- ⊙ A specific time delay between two groups of control signals.
- ⊙ To ensure synchronization, D-Flip Flops are controlled by a common clock signal.

3. Sequence-Counter method:

It uses counter for timing purposes.

4. PLA method

It uses programmable logic array.